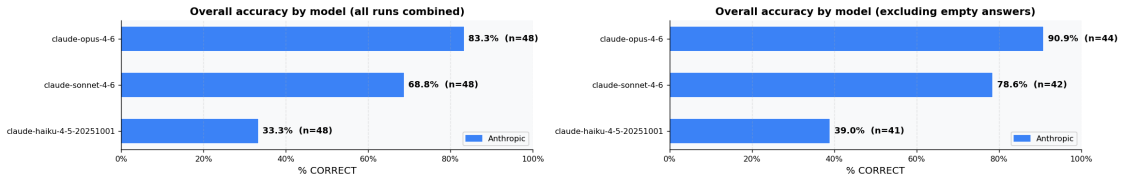
results_analysis_v2

March 14, 2026

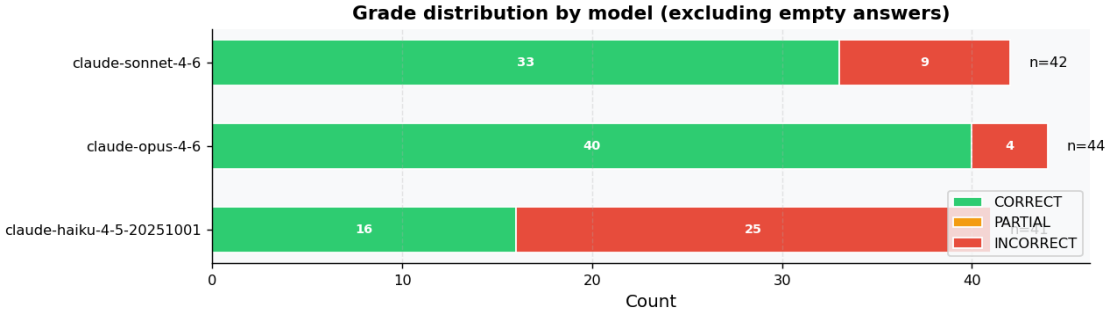
1 TinyBench — Evaluation Results Analysis

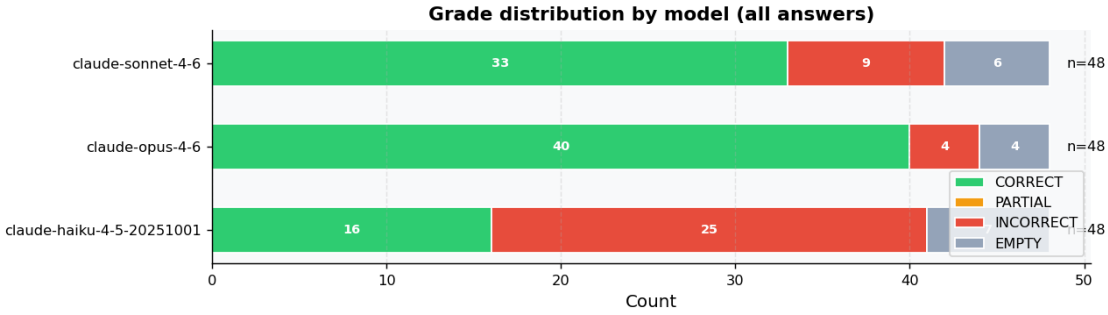
```
Loaded 2 result file(s):
  ../data/results/run_03_13_2026/eval_python_only_graded.json  →
run='python_only'
  ../data/results/run_03_13_2026/eval_python_semantic_graded.json  →
run='semantic'

144 total records  |  3 models  |  2 runs  |  2 templates
Empty answers: 17
Records after excluding empty answers: 127 / 144
Runs present: ['python_only', 'semantic']
```

1.1 1 — Overall Accuracy by Model

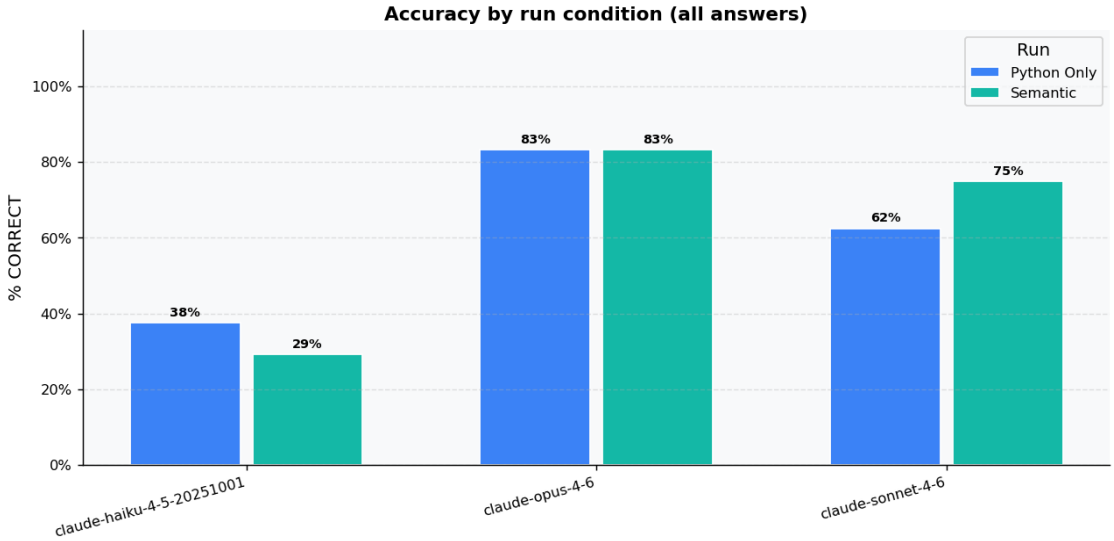


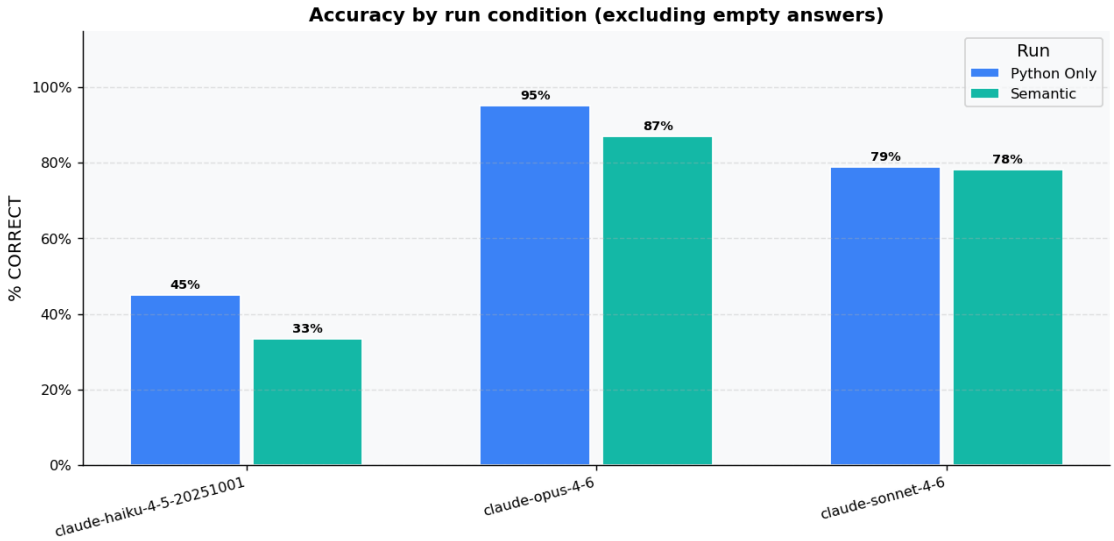
1.2 2 — Grade Distribution

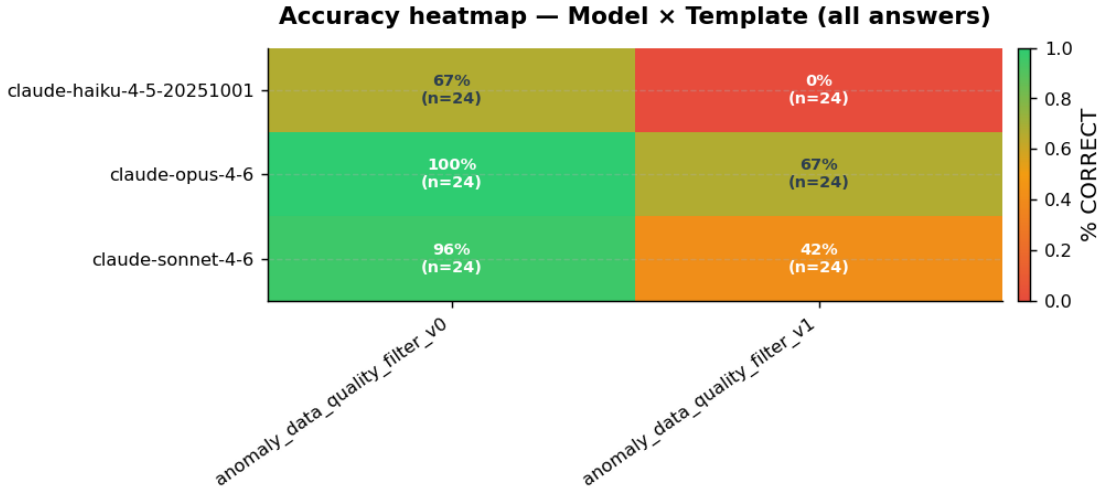


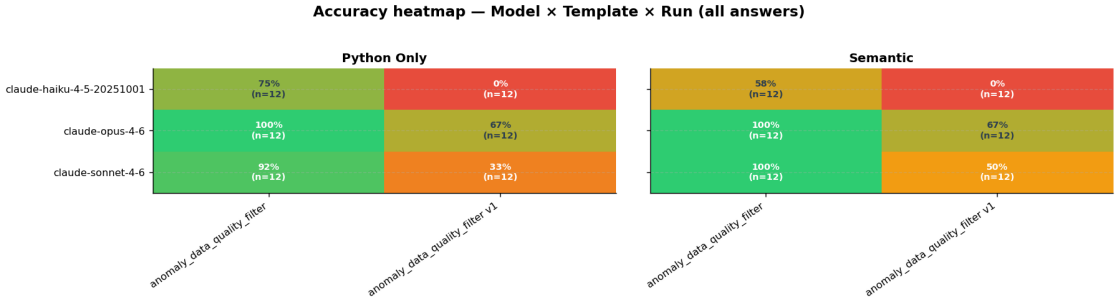


1.3 3 — Accuracy by Run Condition

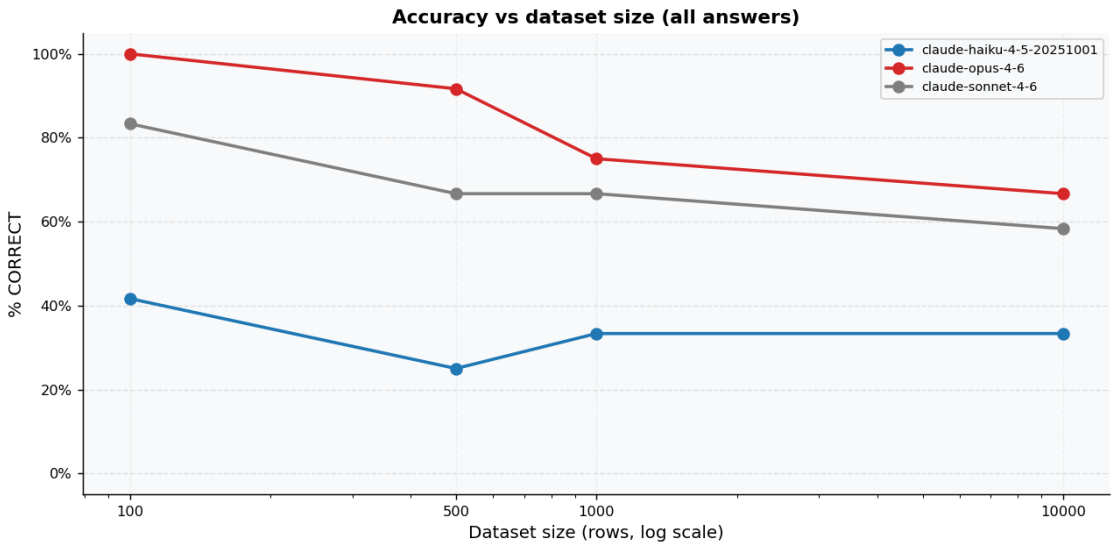


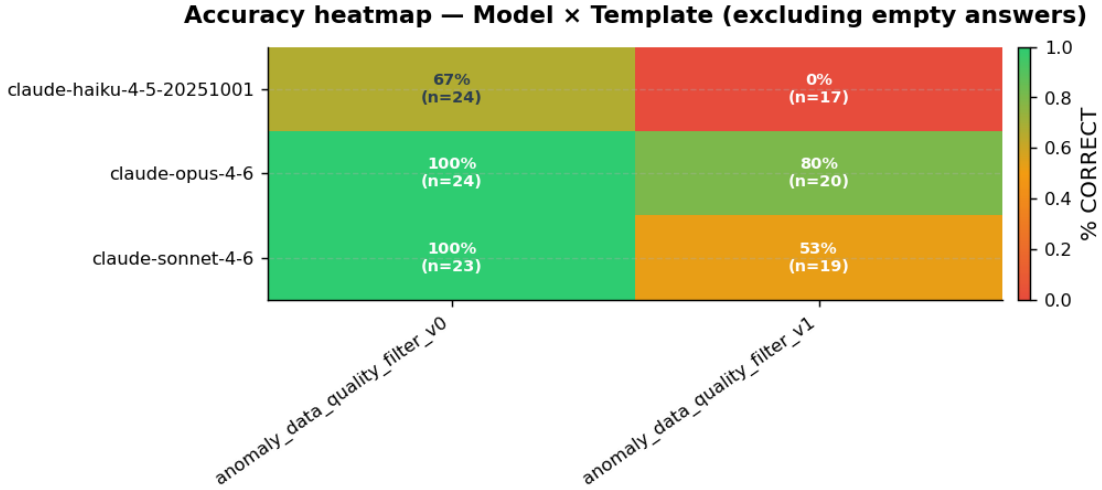

Accuracy by run condition (excluding empty answers)
Run
Python Only
Semantic
% CORRECT
100%
80%
60%
40%
20%
0%
45%
33%
95%
87%
79%
78%
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6

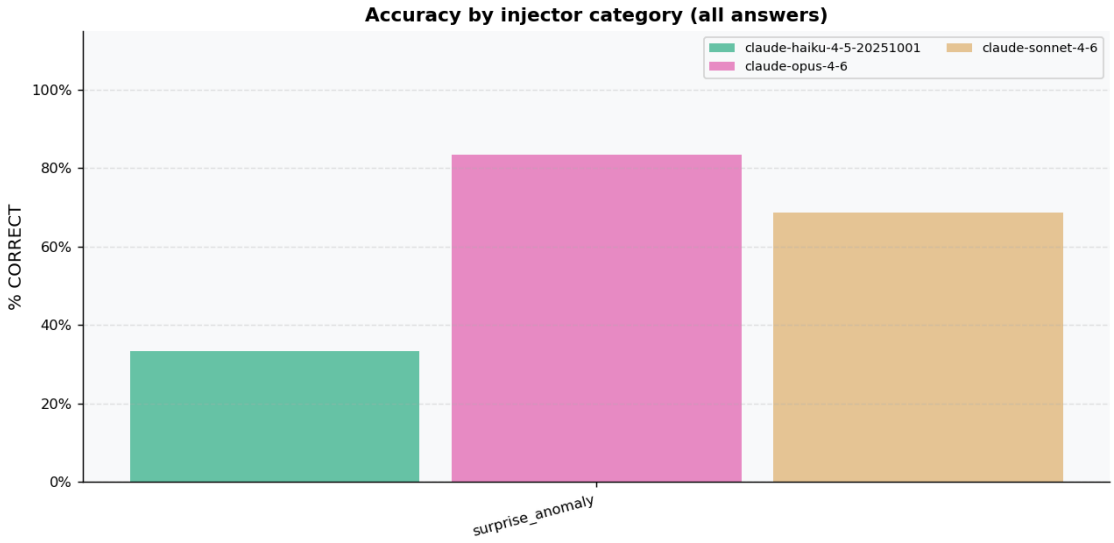

Accuracy heatmap — Model × Template (all answers)
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6
67% (n=24)
0% (n=24)
100% (n=24)
67% (n=24)
96% (n=24)
42% (n=24)
anomaly_data_quality_filter_v0
anomaly_data_quality_filter_v1
% CORRECT
1.0
0.8
0.6
0.4
0.2
0.0


Accuracy heatmap — Model × Template × Run (all answers)
Python Only
Semantic
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6
75% (n=12)
0% (n=12)
100% (n=12)
67% (n=12)
92% (n=12)
33% (n=12)
58% (n=12)
0% (n=12)
100% (n=12)
67% (n=12)
100% (n=12)
50% (n=12)
anomaly_data_quality_filter
anomaly_data_quality_filter_v1
anomaly_data_quality_filter
anomaly_data_quality_filter_v1

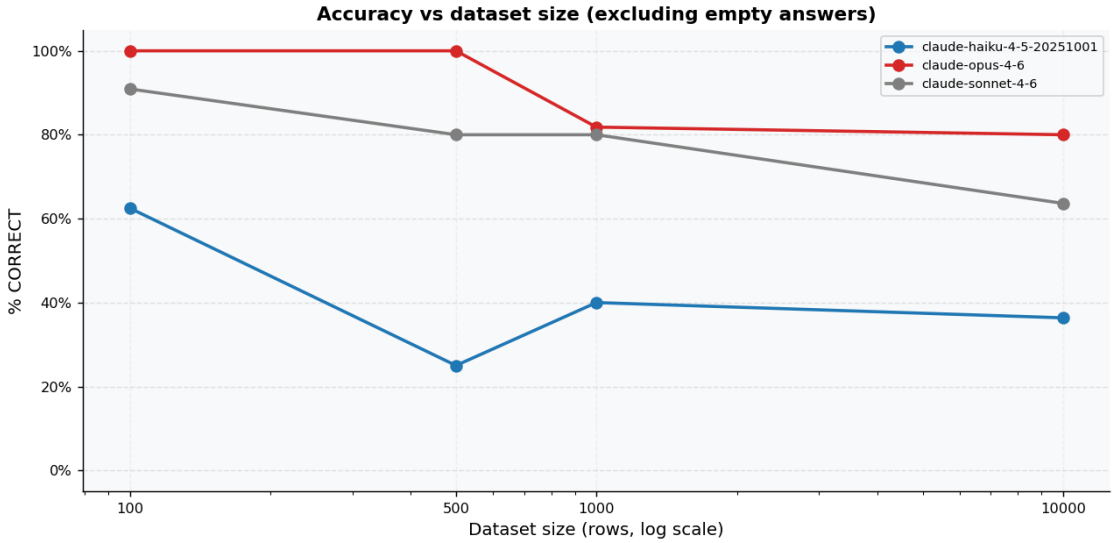
1.4 4 — Accuracy Heatmap: Model × Template



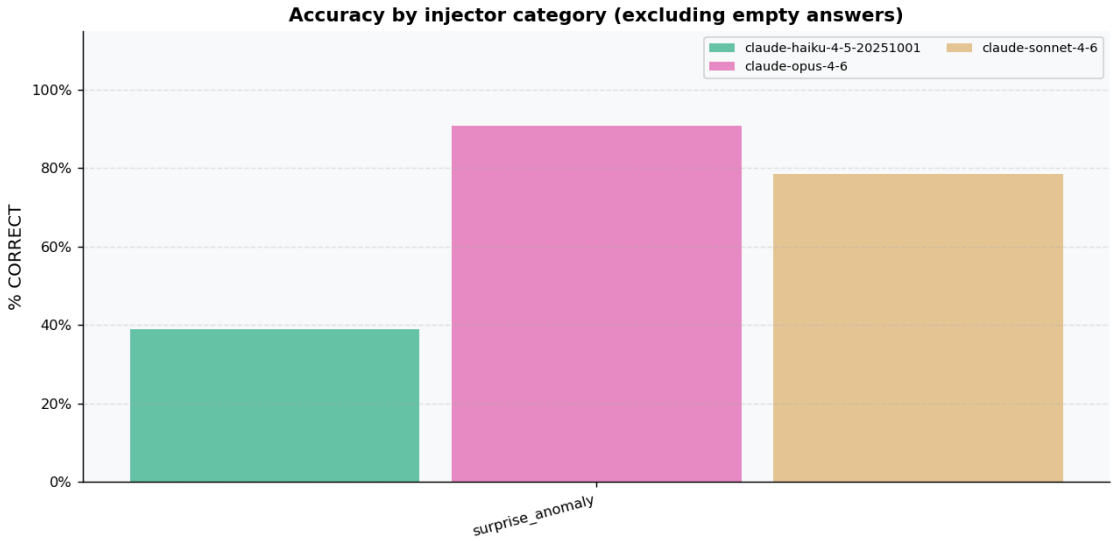




1.5 5 — Accuracy by Dataset Size

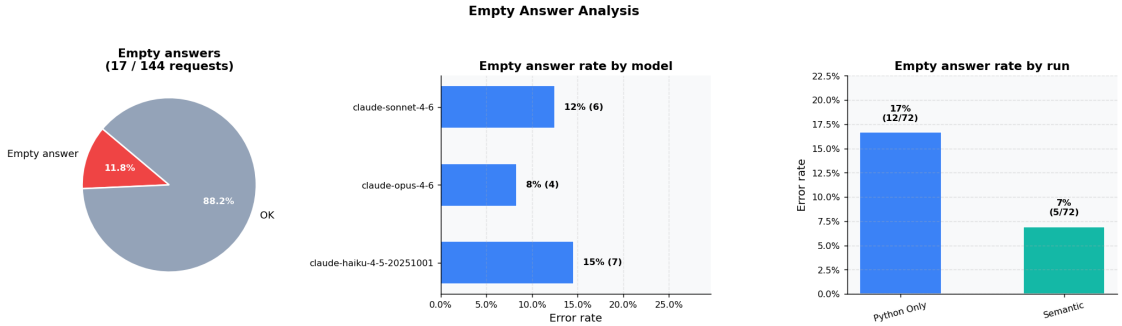


1.6 6 — Accuracy by Injector Category



1.7 7 — Empty Answer Analysis

```
Empty answers: 17 / 144 (11.8%)
Affected datasets: ['bike_sharing_100', 'bike_sharing_1000',
'bike_sharing_10000', 'bike_sharing_500', 'california_housing_100',
'california_housing_1000', 'california_housing_10000', 'california_housing_500']
```



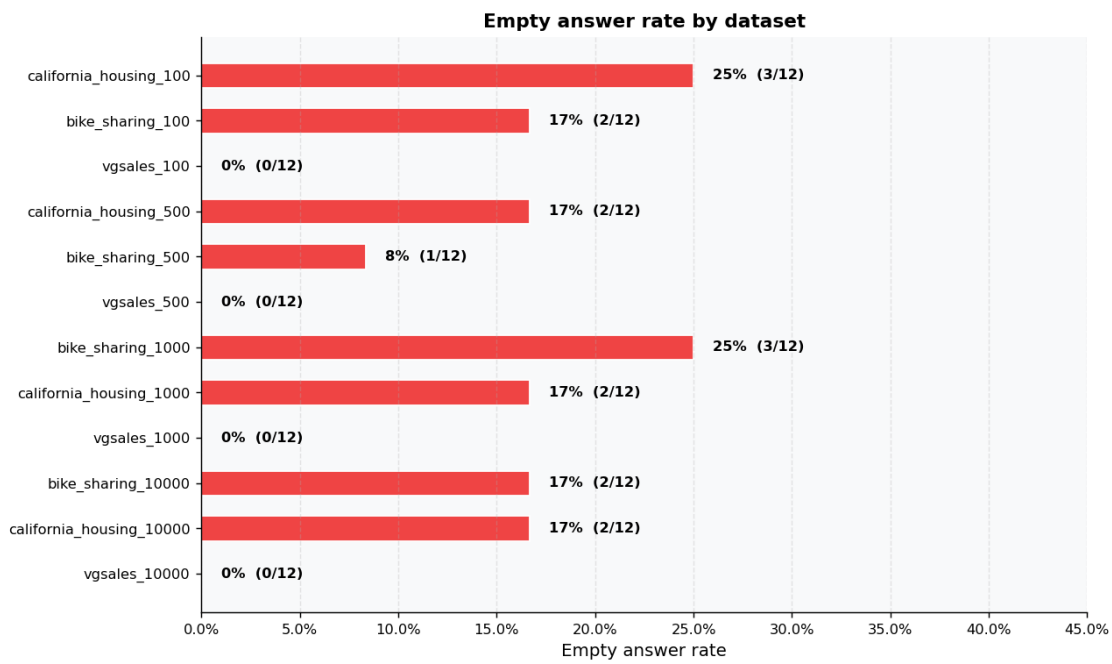

```
Empty answers by model × run:
                                    errors  total rate
model                     run
claude-haiku-4-5-20251001 python_only      4     24  17%
                          semantic         3     24  12%
```

```
claude-opus-4-6          python_only       3     24  12%
                         semantic          1     24   4%
claude-sonnet-4-6        python_only       5     24  21%
                         semantic          1     24   4%
```

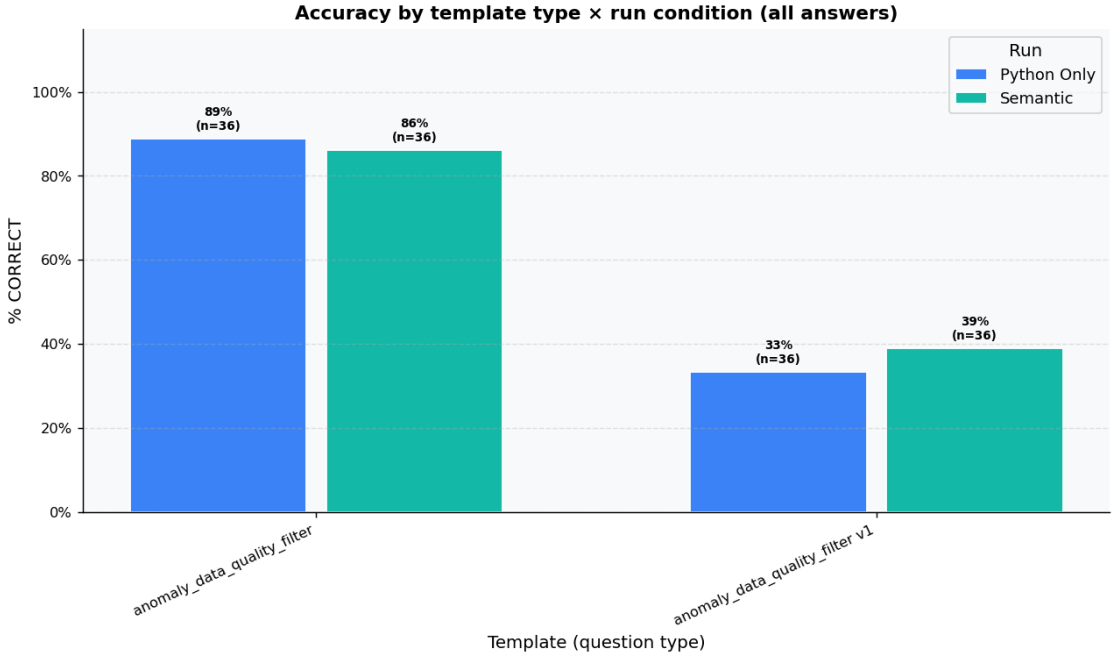
Empty answers by dataset:

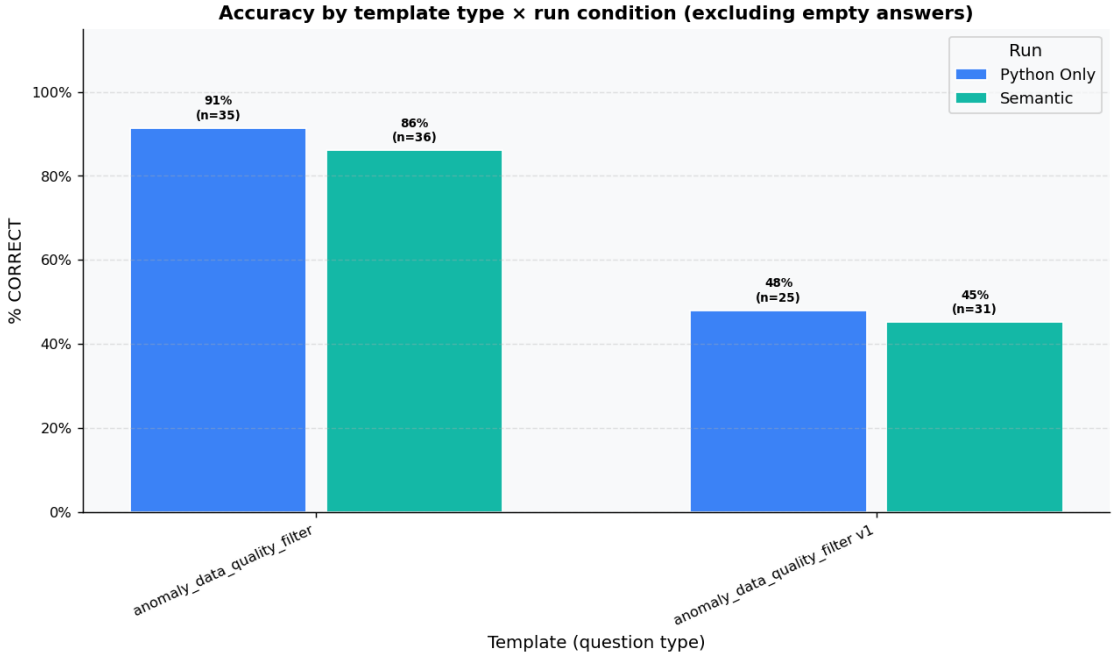
dataset	errors	total	rate	rate_fmt
california_housing_100	3	12	0.25	25%
bike_sharing_100	2	12	0.17	17%
vgsales_100	0	12	0.00	0%
california_housing_500	2	12	0.17	17%
bike_sharing_500	1	12	0.08	8%
vgsales_500	0	12	0.00	0%
bike_sharing_1000	3	12	0.25	25%
california_housing_1000	2	12	0.17	17%
vgsales_1000	0	12	0.00	0%
bike_sharing_10000	2	12	0.17	17%
california_housing_10000	2	12	0.17	17%
vgsales_10000	0	12	0.00	0%

Datasets with errors: 8 / 12



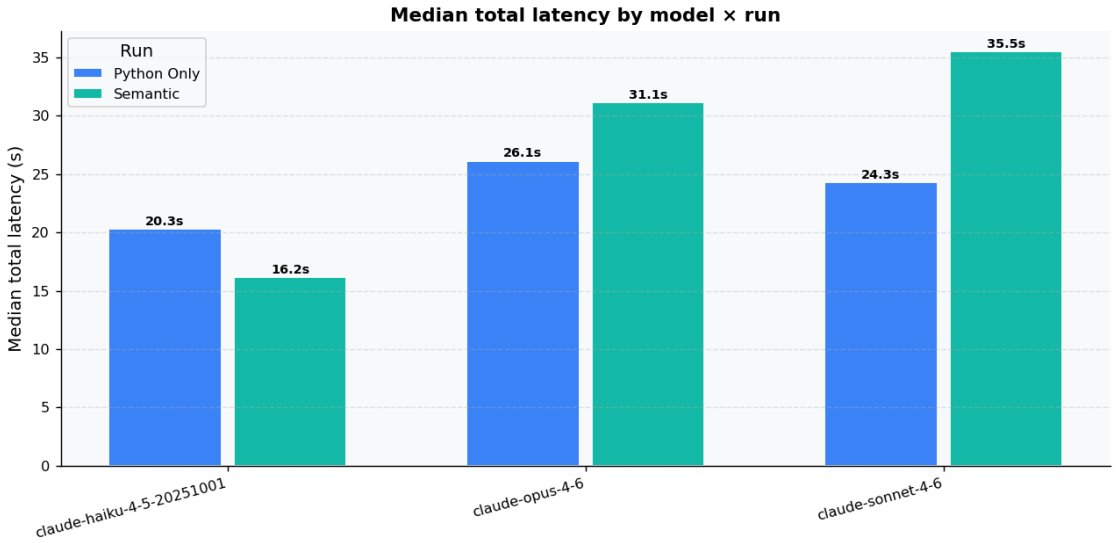
1.8 8 — Template × Run Accuracy



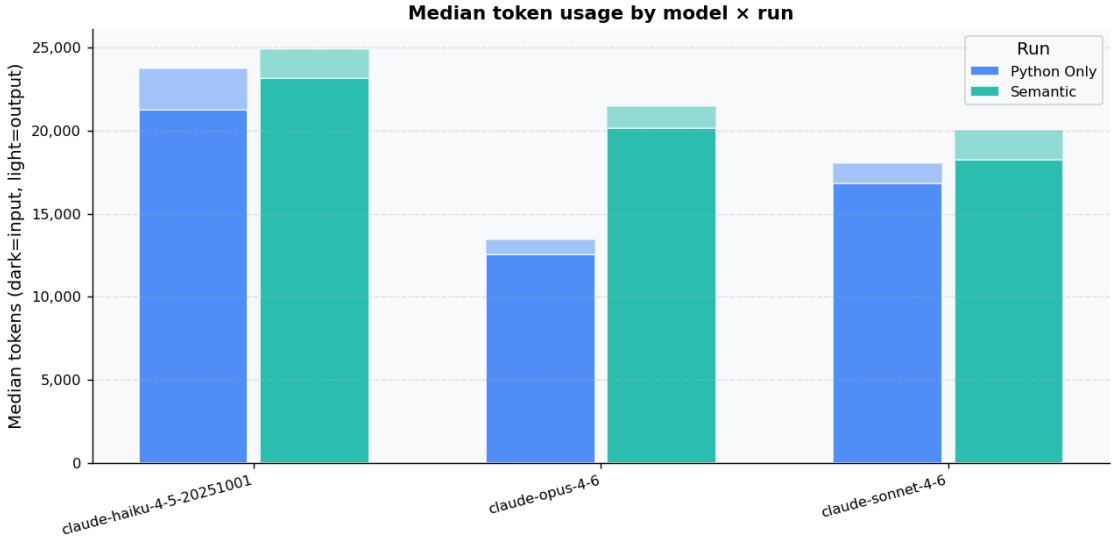


1.9 9 — Latency & Token Metrics

All charts use **median** values (robust to outliers). Error entries are excluded. Metrics: `total_latency_s` (wall-clock), `ttft_s` (time-to-first-token), `input_tokens`, `output_tokens`.



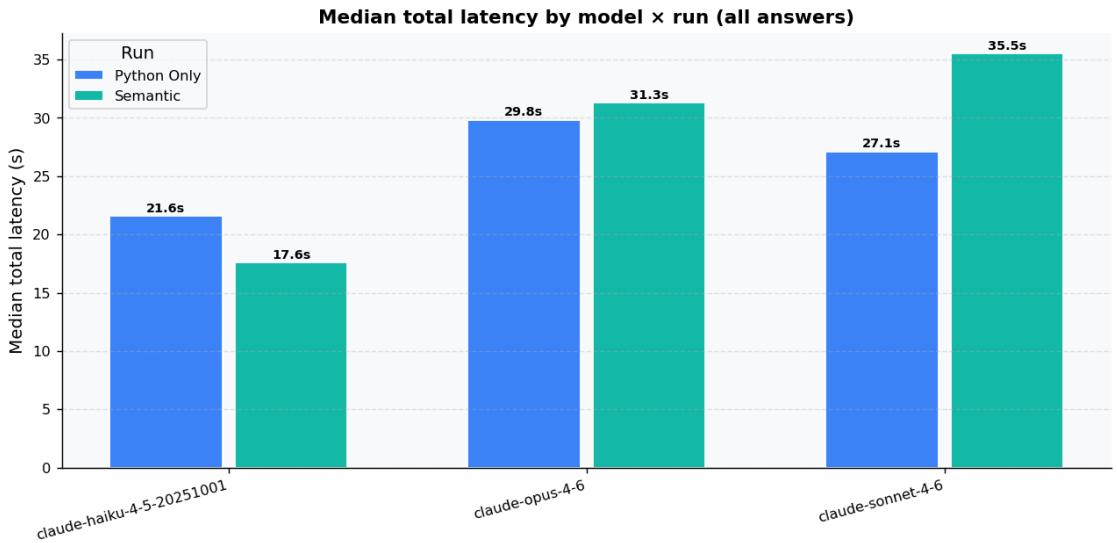
No TTFT data available.

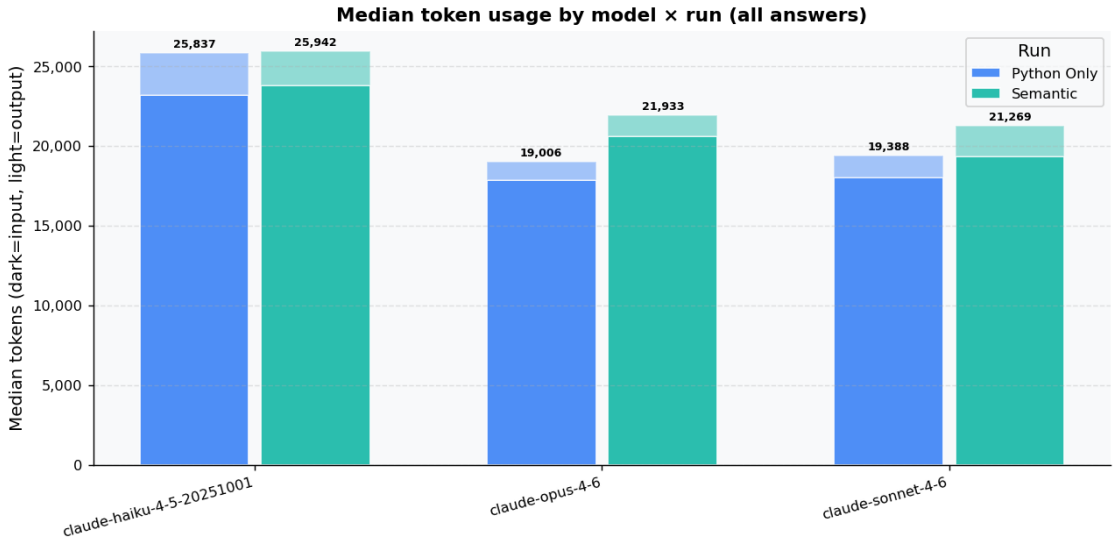


Median metrics per run × model (errors excluded):

```
                                        n  total_latency_s ttft_s  input_tokens
output_tokens  total_tokens
run          model
```

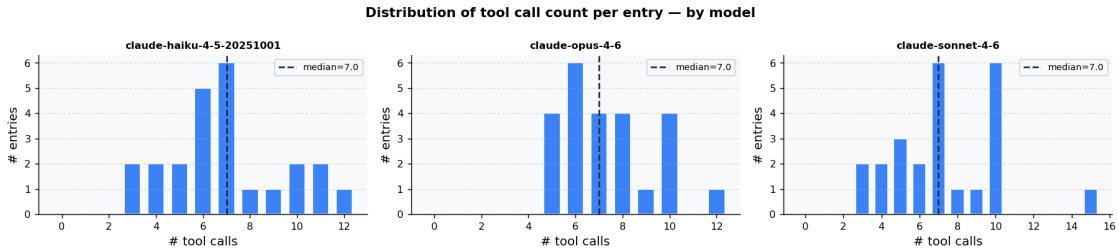
```
Python Only claude-haiku-4-5-20251001  20          20.29   None      21243.50
2488.00      23664.00
           claude-opus-4-6            21          26.13   None      12543.00
920.00     13463.00
           claude-sonnet-4-6          19          24.26   None      16805.00
1229.00      18214.00
Semantic   claude-haiku-4-5-20251001  21          16.17   None      23147.00
1741.00      24757.00
           claude-opus-4-6            23          31.12   None      20151.00
1308.00      21401.00
           claude-sonnet-4-6          23          35.50   None      18274.00
1762.00      20036.00
```

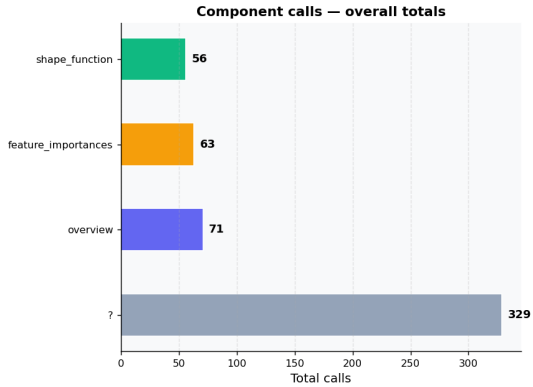


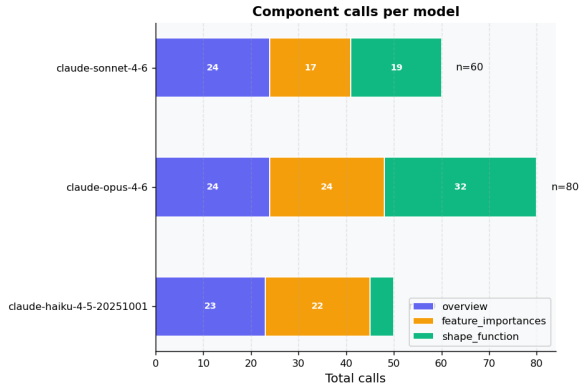


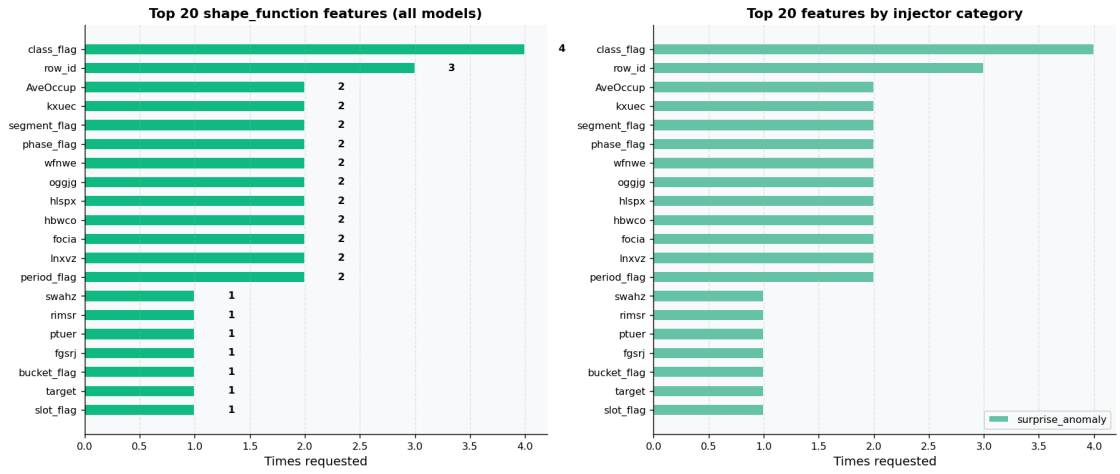
1.10 10 — Semantic Tool Usage Deep Dive

```
Entries in semantic run        : 72
Entries with 0 tool calls      : 0  (0.0%)
Total tool calls made          : 519
Median calls per entry         : 7.0
Max calls per entry            : 15

Calls per entry by model:
                          median  mean  max  zero_calls
model
claude-haiku-4-5-20251001    7.00  6.96   12           0
claude-opus-4-6              7.00  7.38   12           0
claude-sonnet-4-6            7.00  7.29   15           0

Component breakdown (all entries):
component
?                      329
overview                71
feature_importances     63
shape_function          56
```

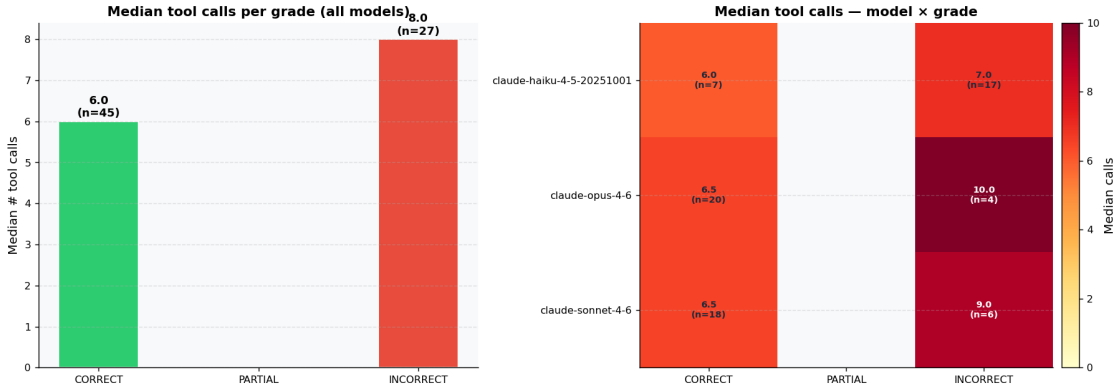








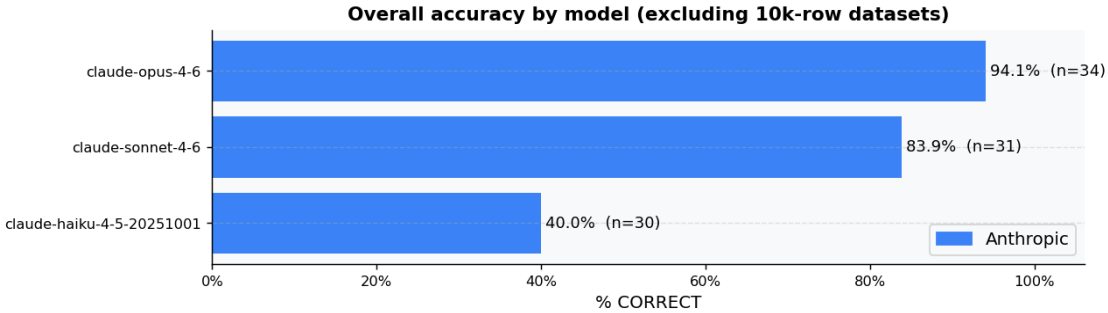
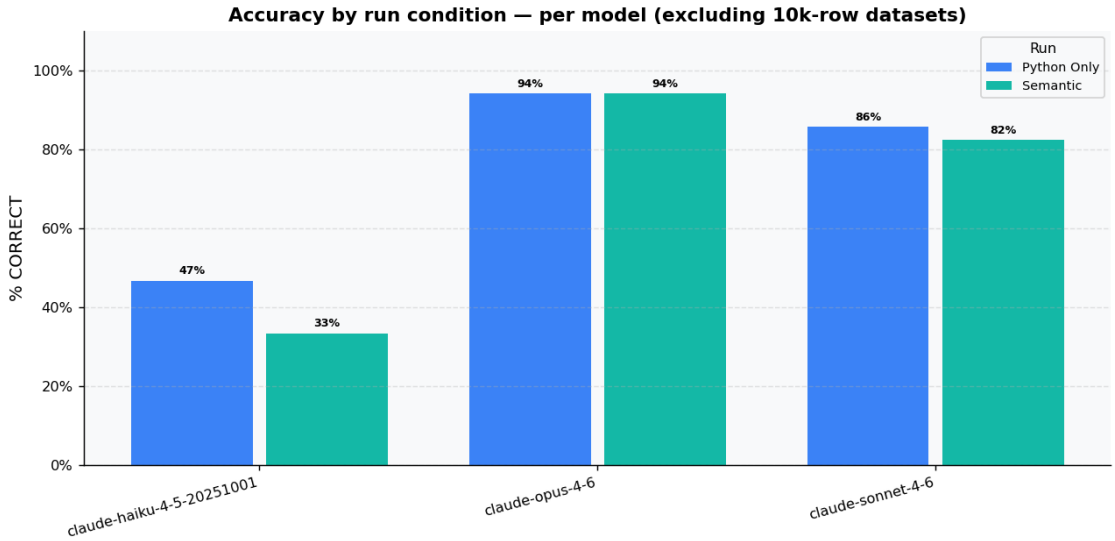

```
posx and posy should be finite values
posx and posy should be finite values
```




```
Entries where no tools were called: 0

No zero-call entries found.
```

1.11 11 — Performance Excluding 10k-Row Datasets

Since 10k-row datasets cause empty answers for csv_context (29% error rate), this section filters them out entirely so we can compare modes on an equal footing.

```
Records after excluding 10k-row datasets: 95 / 127
Datasets kept: ['bike_sharing_100', 'bike_sharing_1000', 'bike_sharing_500',
```

```
'california_housing_100', 'california_housing_1000', 'california_housing_500',
'vgsales_100', 'vgsales_1000', 'vgsales_500']
```